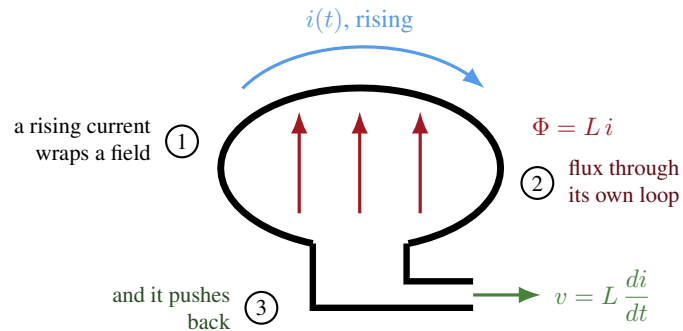
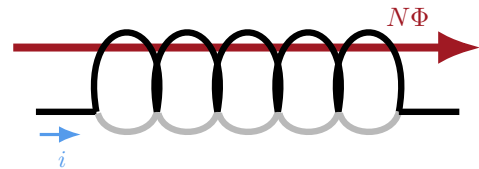


ramp the current, and the loop answers with a voltage

(a) any loop of wire is an inductor



N turns make N times the flux,
and every turn feels all of it:
 $L \propto N^2$

(b) a coil is the same loop, wound N times